UNIVERSITY OF EL OUED
Faculty of Exact Sciences • Department of Computer Science
Second-Year Master: Artificial Intelligence & Data Science (2026/2027)

TP01: Python Web Scraping Case-Study

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1. Objectives & Environment Setup

The objective of this practical work is to implement a robust web scraping pipeline in Python to extract dataset records from an educational website (`books.toscrape.com`) without data duplication, targeting a minimum of 1,000 clean rows, and persisting the final dataset into a CSV file.

Technical Requirements Met:

- **Python 3.x:** Executed via Python environment in VS Code.
- **Requests Library:** Utilized with error handling ('try-except') and timeouts to fetch web pages reliably.
- **BeautifulSoup:** Used for parsing HTML DOM structures and extracting specific product attributes.
- **Pandas:** Structured the collected records into a DataFrame and exported to CSV.

2. Script Architecture & Scraping Logic

The script iterates through pagination links dynamically while checking uniqueness and row counts. Robust exception handling was integrated to handle server connection timeouts past page 36.

Core Logic Snippet:

```
while len(scraped_data) < target_rows:
    url = f"http://books.toscrape.com/catalogue/page-{current_page}.html"
    res = requests.get(url, timeout=10)
    soup = BeautifulSoup(res.text, "html.parser")
    for b in soup.find_all("article", class_="product_pod"):
        title = b.find("h3").find("a")["title"].strip()
        price = b.find("p", class_="price_color").text.strip()
        scraped_data.append({"Book Title": title, "Price": price, ...})
```

3. Execution Results & Schema

The script successfully bypassed network throttling, accumulated exactly 1,000 unique rows across 50 pages, and saved the structured output.

Column Name	Data Description	Extraction Source
`Book Title`	Full title of the book (cleaned string)	`<h3><a title="...">`
`Price`	Product price including currency symbol	`<p class="price_color">`
`Availability`	Stock status (In stock / Out of stock)	`<p class="instock availability">`
`Rating`	Star rating category (e.g., Three, Four)	`<p class="star-rating [Rating]">`

4. Conclusion

The assignment was successfully executed. All core objectives—HTML parsing, robust pagination error handling, dataset compilation, and CSV persistence—were achieved precisely according to specifications in a single-page format.